

Final Project Plan: Unemployment Rates

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DATA 205



Data Sources:

01.

U.S. Bureau of
Labor Statistics

Local Area Unemployment
Statistics
Labor Force Statistics
including the National
Unemployment Rate

Overarching unemployment
data across the U.S. in
percentage rates per year.

02.

State Government
Data

Tennessee
California
Maine
Arkansas
New Jersey

Data per state government
detailing unemployment per
county per year.

03.

Population Dataset:
Census Bureau

Population Estimates and
Projections
Economic Indicators

APIs from the federal
government, population
information and economic
details per year.

Questions

- | | | | |
|----|---|----|---|
| 01 | What trends have affected unemployment rates over time? | 05 | Is there an association between population densities and unemployment rates? |
| 02 | What historical events have affected unemployment rates over time? | 06 | Is there a notable difference in unemployment rates across different geographic areas of the U.S.? |
| 03 | Which counties per state have the highest unemployment rate? | 07 | Are there patterns in historical increases in unemployment? Can they be avoidable? |
| 04 | Is there an association between living conditions and unemployment? | 08 | With our modern socio-political climate, what can we expect out of the future regarding unemployment? |

Methods and processes

Statistical Analysis and Visualizations

- Pearson Correlation test between unemployment and population density
- Interactive GIS map of the U.S. for unemployment distribution
- Summary statistic calculations of unemployment rate (Mean, IQR)
- Diverging bar chart of unemployment rates per state

General Analysis and Research

- Research political shifts, global conflicts, social biases, economic statuses, natural disasters, and urban-rural continuum for the modern day and historic periods
- Look for differences in geographic locations (North/East/South/West)
- Look for repeating patterns in historic events

Risks and Problem Solving

Dates not 100% synced

- Most data is for at least a decade and cut off around 2022 (COVID), there will be more than enough years of consistent overlap
- Turn to the Labor Force Statistics API to collect more information IF NEEDED

Lack of corroborating results

- No result is still a result!
 - Research why unemployment might stabilize, especially if I find times of historical notable crises with no change in unemployment rate to allude to it

Timeline (Moving forward)



April 16th

ALL visualizations
completed and
polished

April 30th

ALL statistical analysis
and research is
complete, begin
polishing and
combining

May 10th

Final report is written
and polished, slides
created and need to be
practiced

May 14th

Final presentation

Thank you

Questions?

Comments?

Concerns?